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ACME PETROCHEMICAL DEMO COMPLEX - CORPORATE ENGINEERING STANDARD

DOCUMENT CODE: STD-ENG-014 | REVISION: R1

TITLE: OVERPRESSURE PROTECTION, PSV SIZING & FLARE HEADER HYDRAULICS (API 520 / API 521 / DIERS)

1. MANDATORY SET PRESSURE & ACCUMULATION LIMITS (ASME SECTION VIII DIV. 1)

- Single PSV (Non-Fire Contingency): Set pressure $\leq$ 100% of authoritative Equipment Internal Design Pressure (MAWP)
- Dual/Multiple PSVs: Primary PSV set pressure $\leq$ 100% of lowest authoritative Equipment Design Pressure; maximum accumulation = 21% above MAWP.
- External Fire Contingency: Maximum accumulation = 21% above MAWP.
- Governance Rule on Conflicting Design Pressures: Where a P&ID and Process Data Sheet exhibit a discrepancy (e.g. P&ID shows a higher design pressure than the Process Data Sheet), the P&ID shall govern.

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STD-ENG-014 PAGE 2 OF 4 - DIERS TWO-PHASE REACTIVE RUNAWAY SIZING

- Reactive Peroxide Systems (Unit 2300 V-2301 & D-2304):

- * Relief valves protecting vessels containing > 5 wt% Cumene Hydroperoxide (CHP) must be sized for homogeneous
- * PSV-2304A/B on D-2304 requires API 526 Orifice 4J6 (8.303 cm²) to pass 38,400 kg/hr two-phase flow at 11.0 kg

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STD-ENG-014 PAGE 3 OF 4 - INLET PRESSURE DROP & BACKPRESSURE LIMITS

- Inlet Piping Pressure Drop: Must not exceed 3.0% of set pressure at rated capacity (requires 6 in short-coupled)
- Balanced Bellows Backpressure Limit: Total superimposed + built-up backpressure in header 8"-PR-2320-316L to D-

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STD-ENG-014 PAGE 4 OF 4 - FLARE KNOCK-OUT DRUM SIZING CRITERIA

- Knock-Out Drum D-2320 must separate liquid droplets ≥ 300 microns and provide 20 minutes of emergency liquid h